

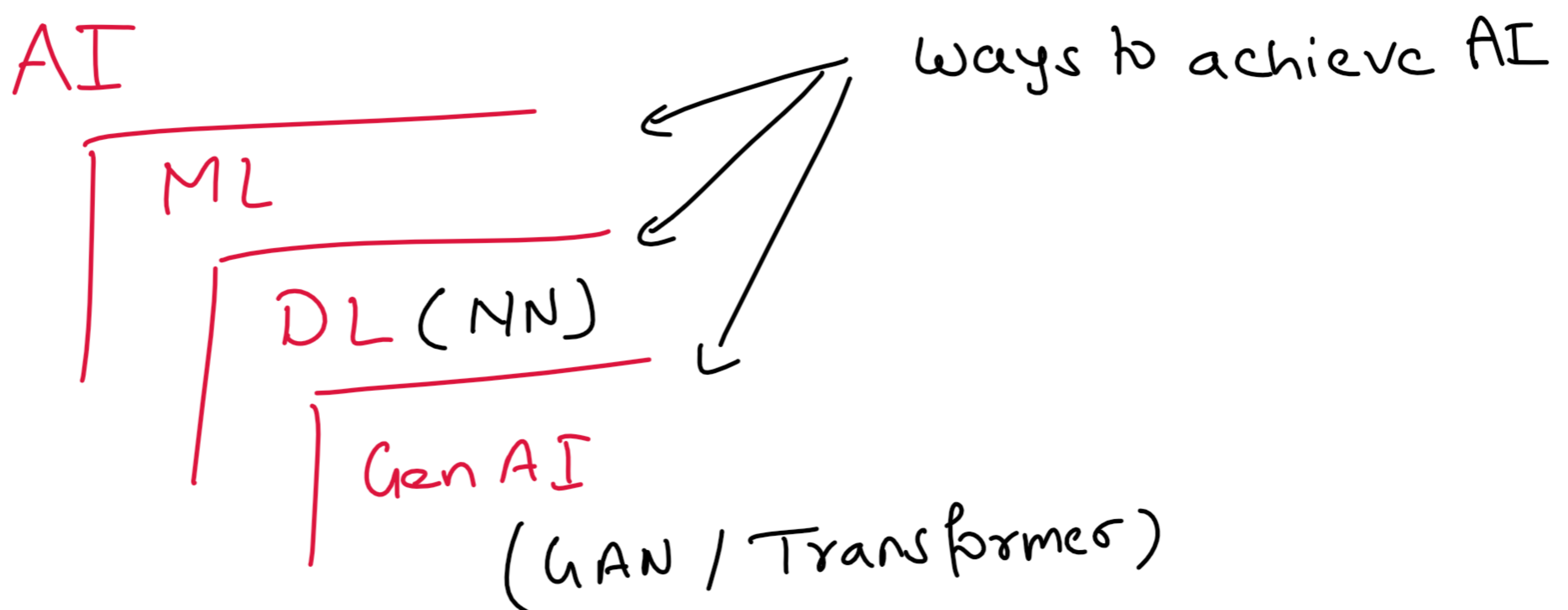
What is AI?

AI : Artificial Intelligence

Traits of human intelligence:

- 1) can easily interpret visual input : image/ video (computer vision)
- 2) can easily interpret text/ spoken (Natural language processing)
- 3) make decision based on past experiences (AI/ ML)
- 4) Generate content (Gen AI)

software/ app can mimic the traits of human intelligence : AI



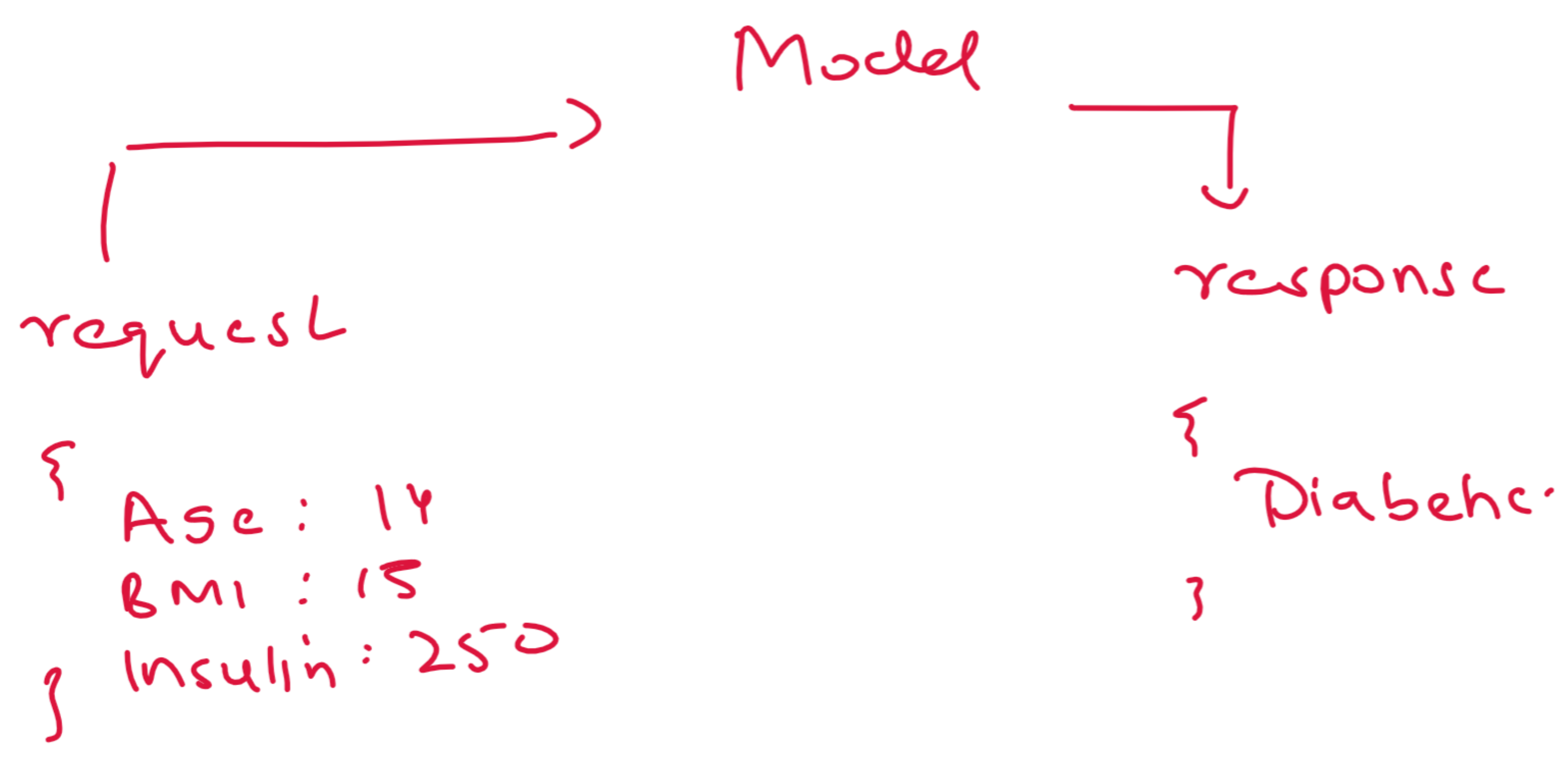
DS → make sense of data
↳ SQL, Excel
↳ ML/AI

Train the model

Machine learning

- **Supervised** → predict label
 - Diabetes (Yes/No)
 - continuous
 - Age (Regression)
- Semi-supervised
- Unsupervised
- Reinforcement

1. Define the problem		input/ features (basis which you want to predict)							
		output/ label (what you want to predict)							
Label:	Predict diabetes (yes/no)								
Features:	Age	BMI	Insulin						
2. Collect the historical data									
	Age	BMI	Insulin	Diabetes					
Train	13	12	100	yes	1				
	14	13	101	yes	1				
	14	102	no	0					
	16	15	103	yes					
	17	16	104	yes					
Test	18		105	no					
	19	18	106	yes					
	20	19	107	yes					
	21	20	108	no					
Preprocessing									
3. Train & Test									
4. Pick algorithm									
Age	BMI	Insulin	=====>	Model	=====>	Diabetes			
						Predicted	Actual		
19	18	106	=====>	Model	=====>	yes	yes	Correct	
20	19	107	=====>	Model	=====>	no	yes	incorrect	
21	20	108	=====>	Model	=====>	no	no	correct	
		Metric	Accuracy					0.666667	
5. Deploy this model (service)									



Azure AI Service : Pre-trained model deployed as a service , custom model
(provide your own data)

-- Integrate gen ai : Azure open ai service

-- Integrate interpret visual input : azure ai vision

-- interpret spoken input : azure speech

--interpret text based input : azure language → Detect language

--translate text : translator service →

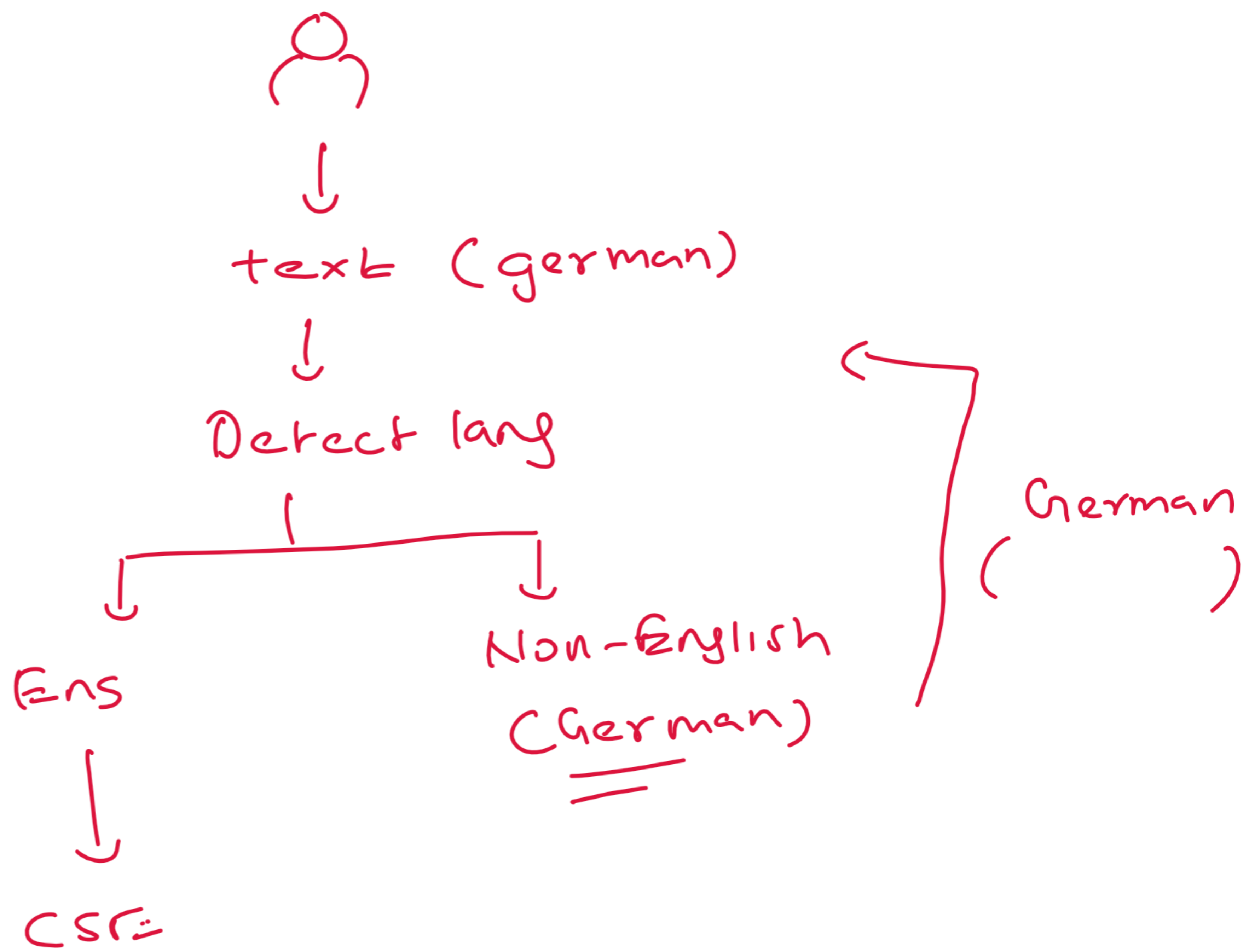
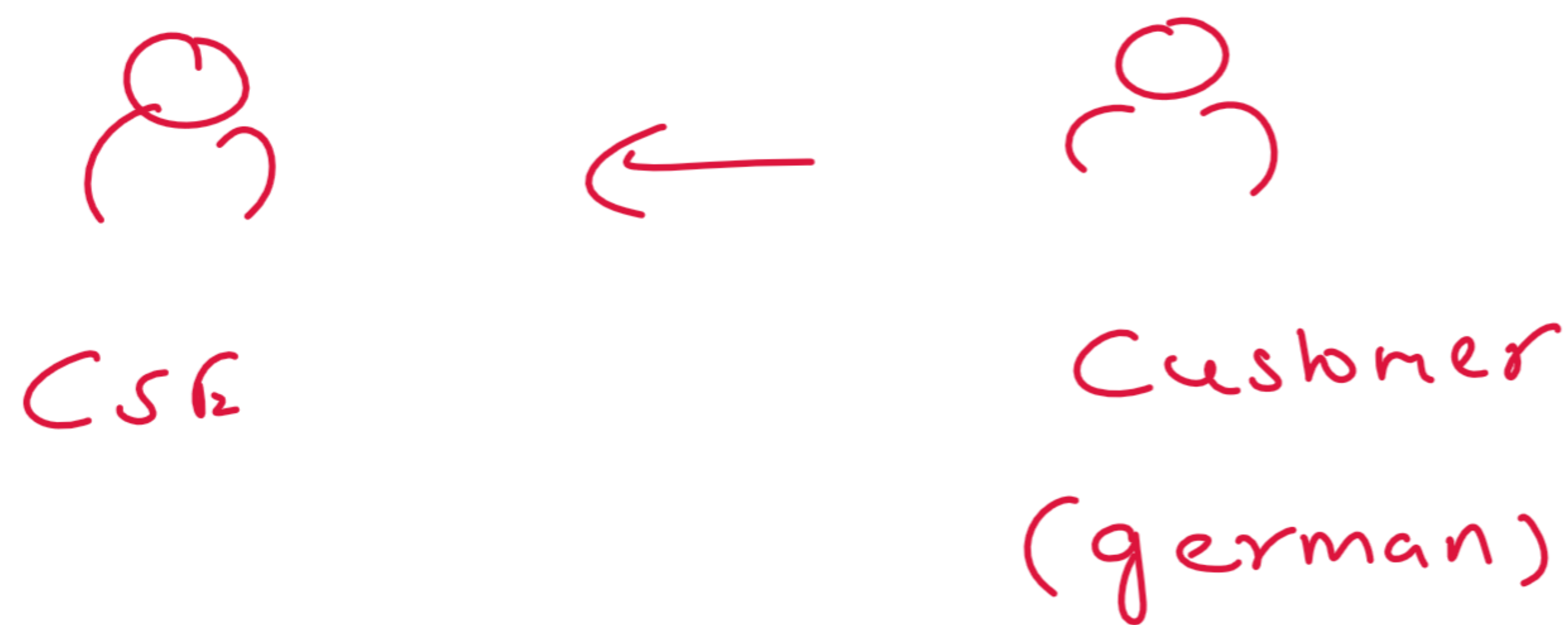
--detect face : face service

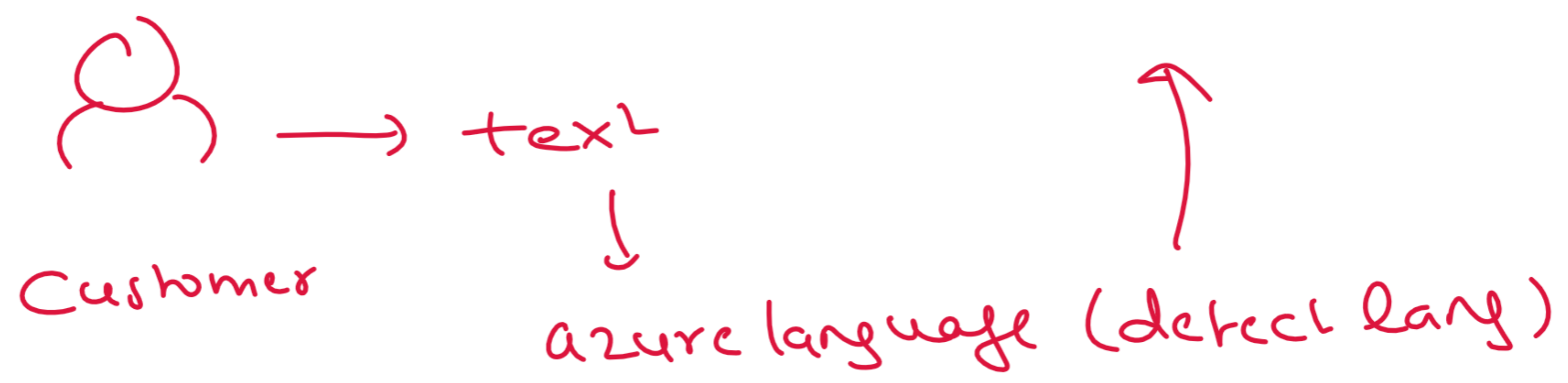
-- custom vision mdel : custom vision

-- search based capability : azure search

Contoso : sells product across multiple geo (native english speakers & non-english speakers)

Customer support (text based , only in english)





~~translator~~

translator



English

Response

Create a resource in azure portal

: Single resource (language service)

-- Free tier

: Multi-resource (language/ vision/ open ai..)

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Software dev toolkit (SDK) (C# & Python)

REST API

consideration of responsible AI

- Fairness: model should be fair to all groups irrespective of their age, ethnicity
- Reliability and Safety :
- Privacy and Security:
- Inclusiveness (accessability)
- Transparency
- Accountability

Symptoms → healthcare
chatbot → prescription

Azure AI foundary

<https://ai.azure.com/>